

# Photoperiodic responses of four wild-trapped desert rodent species

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**El-Bakry, Hanan A., Wafaa M. Zahran, and Timothy J. Bartness.** Photoperiodic responses of four wild-trapped desert rodent species. *Am. J. Physiol.* 275 (*Regulatory Integrative Comp. Physiol.* 44): R2012–R2022, 1998.—The purpose of this study was to examine the effect of the photoperiod on reproductive status and body and lipid masses in four Egyptian desert rodent species (*Dipodillus dasyurus*, *Acomys cahirinus*, *Gerbillus andersoni*, and *Gerbillus pyramidum*). Adult males and females were housed in long days for 11 wk. At that time, one-half of the animals were killed and the remaining animals were moved to short days (SDs) for 11 wk. Some individuals of *Gerbillus andersoni* and *Gerbillus pyramidum* had access to running wheels. Testes index and spermatogenesis, but not testis mass, were decreased in all species in SDs. In contrast, SDs did not affect female reproductive status in all species. Exercise stimulated spermatogenesis but did not affect female reproductive status. SDs increased body and lipid masses in male *Acomys cahirinus*, but not in other species. Collectively, these desert rodent species were responsive to day length changes, but these changes alone did not induce robust alterations in reproductive status and body and lipid masses.

white adipose tissue; gerbils; mice; reproduction; body weight; exercise

SEASONAL CHANGES in the environment can affect a wide range of physiological and behavioral processes in most nontropical mammalian species. For example, dramatic changes in reproductive activity occur across the seasons. Food availability, rainfall, temperature, and day length (i.e., photoperiod) are among the environmental factors that affect the time of onset and the duration of the reproductive season (31). The annual change in day length, however, is the principal environmental cue controlling the timing of reproduction for many nontropical mammals (9,15); other environmental factors serve to shape breeding seasons more subtly. A variety of other seasonal physiological responses often accompany the photoperiod-induced alterations in reproductive status and include changes in pelage (14), body fat (4), and thermogenesis (22). These seasonal responses also are typically regulated by changes in the photoperiod (3).

Seasonal variations in reproductive activity also occur in desert rodents (11, 28, 29, 40, 41, 47). Nevertheless, very little is known about the role of the photoperiod in the control of reproduction or other seasonal responses in desert rodents. The present study ad-

dressed the photoresponsiveness of four Egyptian desert rodent species: Wagner's Dipodids (*Dipodillus dasyurus*), spiny mice (*Acomys cahirinus*), Anderson's gerbils (*Gerbillus andersoni*), and greater gerbils (*Gerbillus pyramidum*). Limited field data indicate the presence of seasonal reproductive cycles in these species (23, 35). Specifically, *Acomys cahirinus* breeds throughout most of the year (February to October), whereas the other three species breed from fall to late spring with a summer hiatus in reproduction (23). The presence of winter breeding in these animals could be due to one of several possibilities. First, these animals could be short day (SD) breeders, such that reproduction is facilitated by SDs and inhibited by long days (LDs). Second, they may have a photoperiodic system much like that of typical LD breeding rodents, but the potential inhibitory effect of SDs is overridden by other environmental factors such as ambient temperature, rainfall, humidity, or food availability. Third, these animals may be reproductively nonresponsive to the photoperiod and instead use other predictors to time their reproduction. Fourth, seasonality in these animals may be a consequence of an opportunistic strategy in which reproduction is restricted to the times when climatic and nutritional conditions are optimal without the use of any environmental predictors. In terms of the last possibility, reproduction in these desert species might occur in SDs when ambient temperatures tend to be mild, rainfall more likely, and food generally more abundant than in the LDs of summer. The purpose of the present experiment was to examine the role of the photoperiod in the control of reproductive status and body and lipid masses in four desert species. Because the availability of running wheels can stimulate the hypothalamic-gonadal axis (27, 38) and interact with the photoperiod (6, 16, 39), we also tested the effects of exercise on reproductive status in LD- and SD-housed animals.

## MATERIALS AND METHODS

### Animals

Adult wild-trapped males and females of four rodent species [*Dipodillus dasyurus* ( $n = 22$ ), *Acomys cahirinus* ( $n = 21$ ), *Gerbillus andersoni* ( $n = 36$ ), *Gerbillus pyramidum* ( $n = 35$ )] were obtained from a commercial supplier (El-Hakim) in August 1996. On arrival, the animals of each species were divided on the basis of gender and group housed (7–10 per cage). They were maintained for 6 wk in LDs ("summerlike"; 14–10-h light-dark cycle; lights on at 0300, lights off at 1700) at a constant temperature ( $22 \pm 2^\circ\text{C}$ ) before the start of the experiment. This LD photoperiod was chosen because it represented the longest day length that occurs at the latitude where the animals live ( $30^\circ\text{N}$ ; Ref. 5). Food (Purina Rodent

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Chow 5001) and tap water were available ad libitum throughout the experiment. All experimental procedures were approved by the Georgia State University Institutional Animal Care and Use Committee, Public Health Services guidelines and also were in accordance with Centers for Disease Control guidelines for housing and handling wild animals (biosafety level 2).

### Experimental Design

**Experiment 1.** All individuals of *Dipodillus dasyurus* and *Acomys cahirinus* were weighed and singly housed in polypropylene cages, each equipped with a running wheel. Locomotor activity was monitored for 11 wk, and body mass was measured weekly to the nearest 0.01 g. After this time, one-half of the animals were anesthetized with methoxyflurane vapors (Metofane, Pitman-Moore, Mundelein, IL) and killed by decapitation, their tissues were harvested, reproductive status was assessed, and carcass composition was measured. The remaining animals were transferred to SDs ("winterlike"; 10:14-h light-dark; lights on at 0600). The length of the SDs represented the shortest day length at the latitude in which the animals were trapped (5). Body mass was measured weekly to the nearest 0.01 g. Eleven weeks later the animals were killed and their tissues were removed and processed as above for the LD-housed animals.

**Experiment 2.** Experiment 2 was designed to test whether exercise affects the reproductive status of LD- or SD-housed animals. Because only a few individuals of *Dipodillus dasyurus* and *Acomys cahirinus* survived trapping and shipping, these two species were not included in this experiment. Individuals of *Gerbillus andersoni* and *Gerbillus pyramidum* were singly housed and divided into two groups balanced for body mass. Animals of the first group were given access to running wheels, whereas the others served as sedentary controls. The same protocol described above for the other two species was followed, except that before being transferred to SDs the males were anesthetized with methoxyflurane and the length and width of the right testis were measured externally through the scrotal skin using handheld calipers. Testicular length and width also were recorded for the males killed in both LDs and SDs. The testicular index was calculated for each animal by multiplying the testis length by testis width. This was not done for the other two species (*Dipodillus dasyurus* and *Acomys cahirinus*) because their testes are located intra-abdominally. Although there was a possibility that these animals were reproductively immature, this was unlikely because several females were pregnant by the time of their arrival.

### Tissue Masses

Various white adipose tissue (WAT) pads, including the inguinal WAT (IWAT), retroperitoneal WAT (RWAT), epididymal WAT (in males), and parametrial WAT (in females) pads were harvested, blotted dry, weighed, and replaced in the carcass.

### Reproductive Status

In all groups, male and female reproductive status was assessed by determining the paired testes or uterine masses, respectively. In addition, testicular indexes were calculated in two species (*Gerbillus andersoni* and *Gerbillus pyramidum*) as above. The testes and ovaries of both species were stored in Bouin's fixative for histological examination of reproductive status (see *Histological Analysis of the Gonads*).

### Carcass Composition

Carcass composition was measured using a modification (2) of the method of Leshner et al. (30). Briefly, the shaved eviscerated carcass was weighed to determine carcass wet weight and dried at 80–85°C to a constant weight to assess carcass water. The dehydrated carcass was blended, lipid was extracted with the use of petroleum ether, and the lipid content was determined gravimetrically. The remaining dehydrated delipidated tissue was termed fat-free dry mass (FFDM).

### Histological Analysis of the Gonads

**Males.** Testes were fixed in Bouin's solution for 2 days, dehydrated, embedded in paraplast, sectioned at 6 µm, and stained with hematoxylin and eosin. The functional state of the testes was determined using the criteria of Grocock and Clarke (18) with some modifications due to the heterogeneity of the testis sections. A spermatogenic index with values 0–5 was applied to 100 randomly selected seminiferous tubules, where 5 is large seminiferous tubules with complete spermatogenesis; 4 is complete spermatogenesis but the number of elongated spermatids and spermatozoa were reduced; 3 is the number of spermatozoa and elongated spermatids further decreased; 2 is only round spermatids found with no elongated spermatids; 1 is only Sertoli cells, spermatogonia, and primary spermatocytes found; 0 is only Sertoli cells present.

**Females.** The ovaries were stored in Bouin's fixative for 24 h, dehydrated, and embedded in paraplast. Serial sections with a thickness of 6 µm were cut and stained with hematoxylin and eosin. The functional state of the ovaries was assessed by examining the number of follicles and corpora lutea in all serial sections throughout one ovary of each animal (36).

### Statistical Analysis

Body mass and spermatogenic index were analyzed with mixed models ANOVA (NCSS 97, NCSS statistical software). Post hoc comparisons of pair-wise means were made using Duncan's multiple-range tests when appropriate. WAT pad, testes and uterine masses, testes indexes, and carcass components were analyzed with a between-subject ANOVA (Systat version 6.0, SPSS). Tukey's honestly significant difference tests were used as post hoc tests when appropriate. Differences between group means were considered statistically significant if  $P < 0.05$ . Test values and exact probabilities are not shown for clarity and simplicity of presentation of the findings.

## RESULTS

### Body Mass

Males had greater body masses than females in *Dipodillus dasyurus*, *Gerbillus andersoni*, and *Gerbillus pyramidum* throughout the study, but this only was statistically significant for *Gerbillus pyramidum* ( $P < 0.05$ ; Figs. 1A and 2). Female *Acomys cahirinus* were heavier than males in all experimental conditions, but these differences were not statistically significant (Fig. 1B). Both LD- and SD-housed *Acomys cahirinus* and *Gerbillus pyramidum* significantly increased their body masses across the experiment ( $P < 0.05$ ; Figs. 1B and 2B) but were not significantly different from one another. There was no difference in body mass between LD- and SD-housed *Dipodillus dasyurus* (Fig. 1A). There was a significant initial decline in body mass in

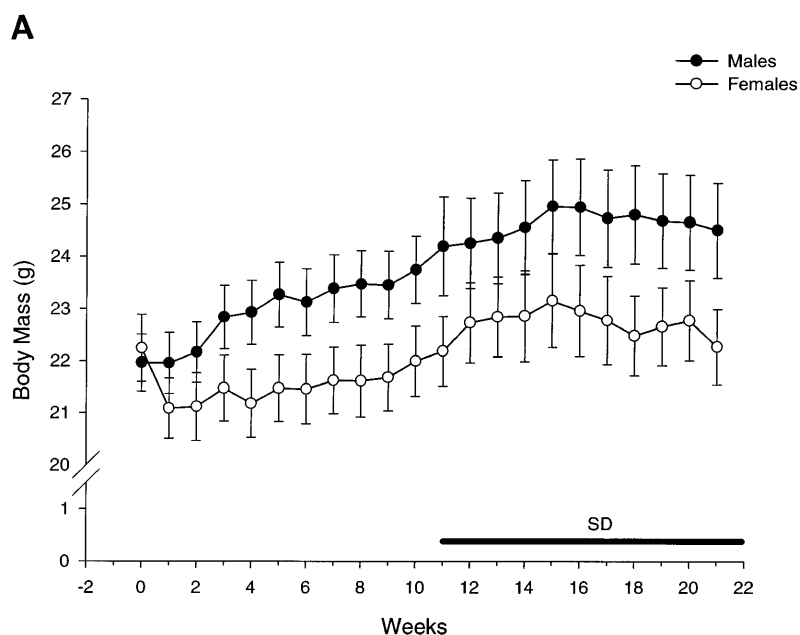
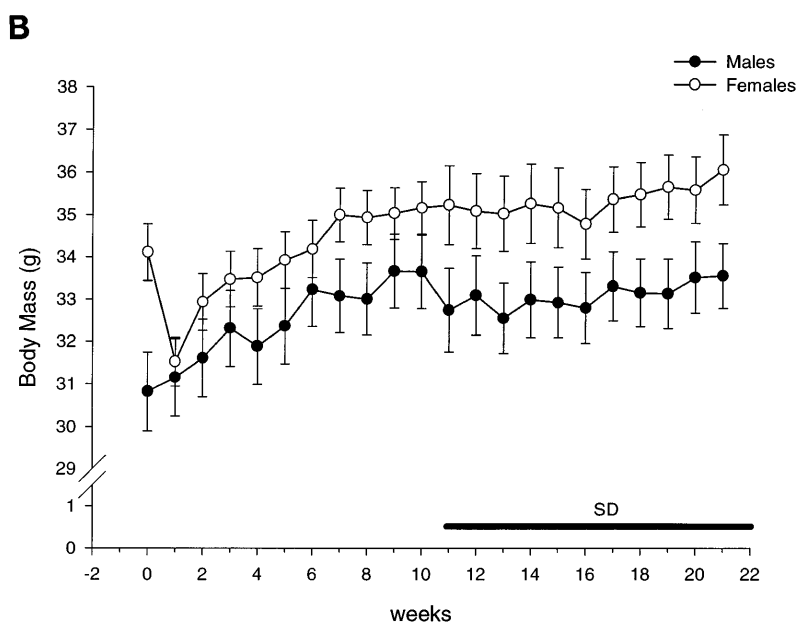


Fig. 1. Mean  $\pm$  SE body mass for male and female *Dipodillus dasyurus* (A) and *Acomys cahirinus* (B). SD, short photoperiod.



the first week of LD exposure in *Gerbillus andersoni* ( $P < 0.05$ ; Fig. 2A) that never recovered fully. Body mass was not affected by photoperiod, exercise, or their interaction in *Gerbillus andersoni* or *Gerbillus pyramidum* (Fig. 2).

#### Fat Pad Masses

IWAT and RWAT masses were increased in male *Gerbillus pyramidum* compared with both LD- and SD-housed females ( $P < 0.05$ ; Table 1). Fat pad masses were not different between LD- and SD-housed animals of all species studied except for two exceptions. First, the IWAT mass of SD-housed male *Acomys cahirinus* was greater than that of their LD-housed counterparts ( $P < 0.05$ ; Table 1). Second, the photoperiod, combined

with the access to an exercise wheel, resulted in greater IWAT mass in SD- than LD-housed *Gerbillus andersoni* ( $P < 0.05$ ; Table 1).

#### Carcass Composition

Two gender effects were found among these species for the carcass composition measures. First, all carcass components were greater in male *Gerbillus pyramidum* than females in both photoperiods for absolute ( $P < 0.05$ ; Table 2), but not relative (corrected for body masses), measures. Second, carcass lipid content was greater in female *Acomys cahirinus* than in males in both photoperiods ( $P < 0.05$ ; Table 2). SD exposure increased absolute and relative carcass lipid content in male *Acomys cahirinus* ( $P < 0.05$ ; Table 2). Finally,

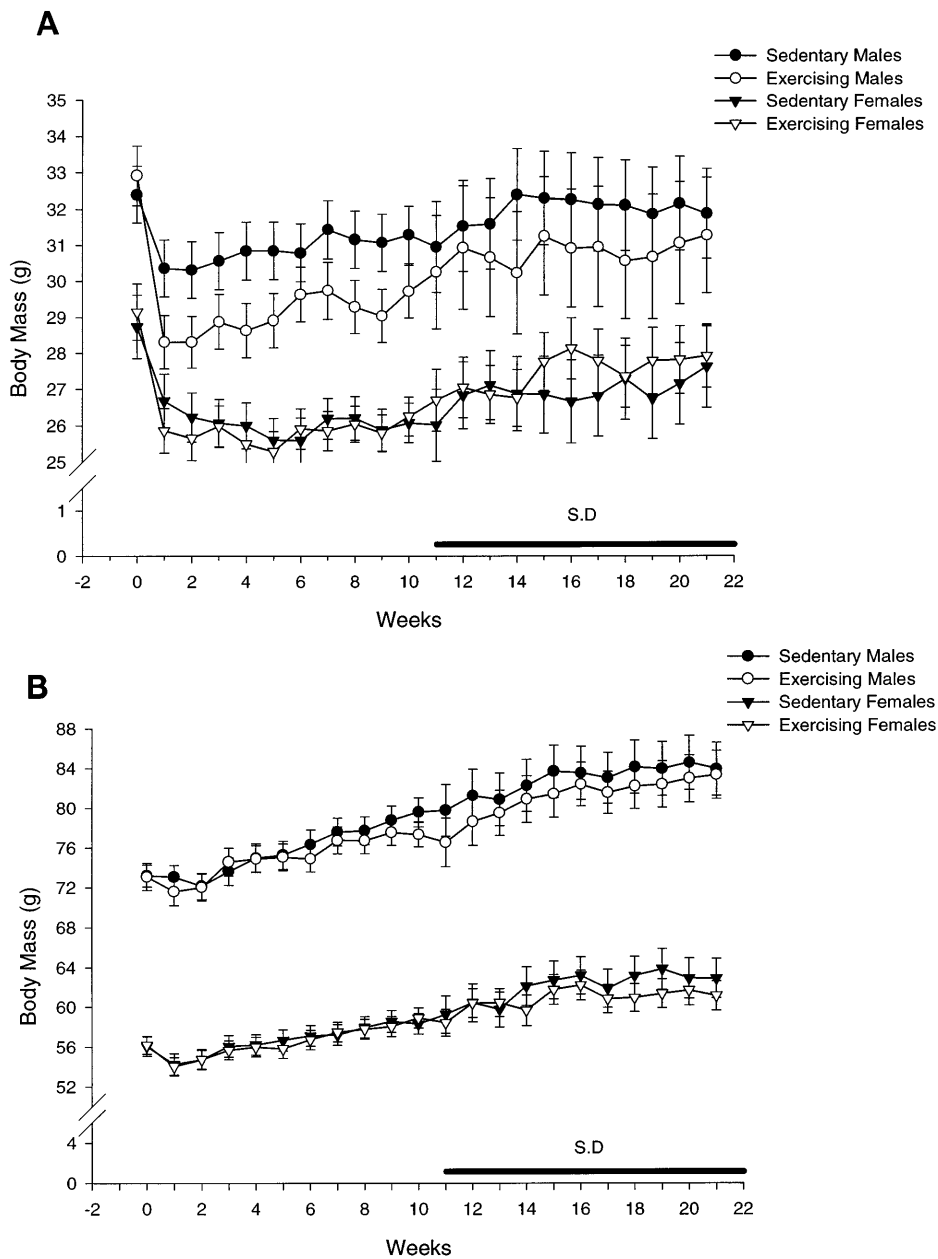


Fig. 2. Mean  $\pm$  SE body mass for male and female *Gerbillus andersoni* (A) and *Gerbillus pyramidum* (B). Sedentary, animals housed in conventional cages; Exercising, animals given access to running wheels.

exercise did not affect carcass composition, except for lipid and FFDM responses in SD-housed *Gerbillus andersoni*. Specifically, lipid content increased and FFDM decreased in SD-housed wheel running animals compared with their sedentary counterparts ( $P < 0.05$ ; Table 2).

#### Reproductive Status

**Males.** Neither absolute nor relative testes masses differed significantly between LD- and SD-housed males of any species (Table 3). The testes of both LD- and SD-housed *Dipodillus dasyurus* had very low numbers of seminiferous tubules (i.e.,  $<100$ ). Therefore, this species was excluded from further statistical analysis of spermatogenic activity, although the testicular tissues of all animals were examined histologically and the same criteria for judging the spermatogenic activity

in all other species were followed. Testes of LD-housed male *Acomys cahirinus* and *Dipodillus dasyurus* were active spermatogenically, and the majority of the seminiferous tubules were in complete spermatogenesis with large numbers of spermatozoa in their lumens (index 5; Figs. 3A and 4A). Eleven weeks of SD exposure caused a pronounced decrease in the spermatogenic activity, but none of the males underwent complete gonadal regression. Specifically, the number of seminiferous tubules with spermatogenic index of 5 significantly decreased, whereas the number of seminiferous tubules with spermatogenic index of 4, 3, 2, and 1 significantly increased ( $P < 0.05$ ; Figs. 3A and 4B).

Neither exercise nor photoperiod affected absolute or relative testes masses in *Gerbillus pyramidum* and *Gerbillus andersoni* (Table 3). In contrast, photoperiod, but not exercise, significantly affected testicular index

Table 1. Effect of photoperiod on WAT masses

| Group                            | Males       |              |             |             |             |             |
|----------------------------------|-------------|--------------|-------------|-------------|-------------|-------------|
|                                  | IWAT        |              | RWAT        |             | EWAT        |             |
|                                  | LD          | SD           | LD          | SD          | LD          | SD          |
| <i>Dipodillus dasyurus</i>       | 0.32 ± 0.07 | 0.51 ± 0.14  | 0.07 ± 0.04 | 0.1 ± 0.01  | 0.12 ± 0.06 | 0.14 ± 0.04 |
| <i>Acomys caharinus</i>          | 0.31 ± 0.05 | 0.51 ± 0.03* | 0.05 ± 0.01 | 0.1 ± 0.02  | 0.17 ± 0.03 | 0.3 ± 0.08  |
| <i>Gerbillus pyramidum</i> ; WR  | 0.73 ± 0.13 | 0.62 ± 0.06  | 0.22 ± 0.08 | 0.19 ± 0.06 | 0.76 ± 0.24 | 0.81 ± 0.16 |
| <i>Gerbillus pyramidum</i> ; Sed | 0.64 ± 0.26 | 0.91 ± 0.36  | 0.33 ± 0.07 | 0.24 ± 0.02 | 1.06 ± 0.24 | 1.10 ± 0.35 |
| <i>Gerbillus andersoni</i> ; WR  | 0.33 ± 0.08 | 0.28 ± 0.12  | 0.04 ± 0.01 | 0.05 ± 0.01 | 0.28 ± 0.09 | 0.24 ± 0.04 |
| <i>Gerbillus andersoni</i> ; Sed | 0.50 ± 0.12 | 0.61 ± 0.19† | 0.08 ± 0.02 | 0.08 ± 0.03 | 0.38 ± 0.04 | 0.40 ± 0.1  |

| Group                            | Females     |              |             |             |              |              |
|----------------------------------|-------------|--------------|-------------|-------------|--------------|--------------|
|                                  | IWAT        |              | RWAT        |             | PWAT         |              |
|                                  | LD          | SD           | LD          | SD          | LD           | SD           |
| <i>Dipodillus dasyurus</i>       | 0.24 ± 0.08 | 0.26 ± 0.04  | 0.05 ± 0.03 | 0.06 ± 0.02 | 0.03 ± 0.01  | 0.04 ± 0.02  |
| <i>Acomys caharinus</i>          | 0.35 ± 0.03 | 0.45 ± 0.03  | 0.06 ± 0.02 | 0.08 ± 0.02 | 0.07 ± 0.01  | 0.08 ± 0.02  |
| <i>Gerbillus pyramidum</i> ; WR  | 0.40 ± 0.06 | 0.37 ± 0.04  | 0.14 ± 0.03 | 0.14 ± 0.02 | 0.05 ± 0.01  | 0.06 ± 0.01  |
| <i>Gerbillus pyramidum</i> ; Sed | 0.39 ± 0.07 | 0.46 ± 0.06  | 0.10 ± 0.01 | 0.18 ± 0.06 | 0.09 ± 0.01  | 0.06 ± 0.03  |
| <i>Gerbillus andersoni</i> ; WR  | 0.51 ± 0.26 | 0.18 ± 0.04  | 0.05 ± 0.03 | 0.03 ± 0.01 | 0.03 ± 0.01  | 0.03 ± 0.01  |
| <i>Gerbillus andersoni</i> ; Sed | 0.27 ± 0.05 | 0.33 ± 0.07† | 0.03 ± 0.01 | 0.06 ± 0.02 | 0.04 ± 0.003 | 0.04 ± 0.001 |

Data represent means ± SE. LD, long photoperiod (14:10-h light-dark cycle); SD, short photoperiod (10:14-h light-dark cycle); IWAT, inguinal white adipose tissue; RWAT, retroperitoneal white adipose tissue; EWAT, epididymal white adipose tissue; PWAT, parametrial white adipose tissue; WR, wheel running animals; Sed, sedentary animals. \* $P < 0.05$  vs. LD-housed animals. † $P < 0.05$  vs. SD-housed WR animals.

(Table 4). Specifically, testicular indexes were significantly lower in SD- compared with LD-housed animals ( $P < 0.05$ ; Table 4). These results also were found when comparing testicular indexes for the same animals before and after transferring them to SDs (Table 4). Photoperiod and exercise conditions did not interact to affect the testicular indexes in any species.

Spermatogenic activity in the seminiferous tubules was markedly decreased in SD-housed *Gerbillus pyramidum* and *Gerbillus andersoni* compared with LD-housed animals ( $P < 0.05$ ; Fig. 3, B and C). In addition, there was a significant effect of exercise on spermatogenic activity. Specifically, sedentary animals exhibited less spermatogenic activity compared with wheel running animals in that sedentary animals had a significantly decreased number of seminiferous tubules with spermatogenic index of 5 and an increased number of seminiferous tubules with spermatogenic index of 4, 3, 2, and 0 (Fig. 3, B and C). Photoperiod and exercise also did not interact to affect spermatogenic activity in any species. There was a clear difference in the response of these two *Gerbillus* species to SD exposure. That is, SD-housed *Gerbillus andersoni* had two distinct testicular responses. First, spermatogenic activity was less than that of the LD-housed animals, but the difference was moderate despite being statistically significant. Specifically, the seminiferous tubules had all stages of spermatogenesis, but the number of spermatozoa and elongated spermatids dramatically decreased (Fig. 5, A and B). Second, spermatogenesis was inhibited to a great extent associated with a pronounced testicular regression. Specifically, spermatogenesis was arrested at the early stages with absence of spermatozoa as well as elongated and round spermatids and presence of necrotic cells in many of the seminiferous tubules (Fig. 5C). The diameter of the seminiferous tubules also was

greatly reduced (Fig. 5C). The latter pattern was found in sedentary animals and was seen in 25% of exercising *Gerbillus andersoni*. In contrast, although male *Gerbillus pyramidum* had moderate, but significant, changes in spermatogenic activity in which the number of spermatozoa and spermatids greatly decreased in SDs, there was no evidence of complete testicular regression (Fig. 6, A and B).

**Females.** Photoperiod did not affect the female reproductive status of any of the species. That is, uterine masses did not differ significantly between LD- and SD-exposed animals (Table 3). Exercise by itself, or in conjunction with SD exposure, also did not influence the uterine masses (Table 3). These results were confirmed by the histological examination of the ovaries. Specifically, females of all species had all stages of folliculogenesis in both LDs and SDs (data not shown).

## DISCUSSION

On the basis of the limited field data available for these and related desert species, we predicted that short photoperiods would stimulate and long photoperiods would inhibit reproductive status; however, this hypothesis was not supported. Changes in the photoperiod affected reproductive status only in males of the four desert rodent species studied (i.e., *Acomys caharinus*, *Dipodillus dasyurus*, *Gerbillus pyramidum*, and *Gerbillus andersoni*), albeit somewhat subtly and then opposite to what we predicted. That is, SD exposure decreased testes index and spermatogenesis (as indicated by a reduced spermatid index) but did not affect testes wet masses. The photoperiod did not affect reproductive status for females in any of the species. Specifically, the ovaries of all species had complete ovarian folliculogenesis in both photoperiods. Exercise

Table 2. Effect of photoperiod treatment on carcass composition

| Group                            | Carcass Component         |              |               |              |              |               |
|----------------------------------|---------------------------|--------------|---------------|--------------|--------------|---------------|
|                                  | Total CW, g               |              | Total FFDM, g |              | Total BF, g  |               |
|                                  | LD                        | SD           | LD            | SD           | LD           | SD            |
| <i>Males</i>                     |                           |              |               |              |              |               |
| <i>Dipodillus dasyurus</i>       | 9.50 ± 0.28               | 9.26 ± 0.50  | 3.65 ± 0.25   | 3.77 ± 0.18  | 1.72 ± 0.71  | 3.76 ± 1.04   |
| <i>Acomys caharinus</i>          | 14.57 ± 1.08              | 14.18 ± 0.62 | 5.46 ± 0.34   | 5.81 ± 0.19  | 1.91 ± 0.33  | 3.56 ± 0.16*  |
| <i>Gerbillus pyramidum</i> ; WR  | 33.64 ± 1.11              | 39.2 ± 2.60  | 13.27 ± 0.23  | 15.18 ± 1.24 | 6.4 ± 1.95   | 6.5 ± 1.45    |
| <i>Gerbillus pyramidum</i> ; Sed | 34.09 ± 3.63              | 37.27 ± 3.81 | 14.03 ± 1.2   | 14.67 ± 0.57 | 6.72 ± 0.80  | 9.62 ± 3.75   |
| <i>Gerbillus andersoni</i> ; WR  | 11.22 ± 0.55              | 12.45 ± 1.09 | 4.56 ± 0.37   | 4.67 ± 0.39  | 2.96 ± 0.92  | 1.26 ± 1.16   |
| <i>Gerbillus andersoni</i> ; Sed | 11.78 ± 0.68              | 11.06 ± 0.39 | 4.64 ± 0.25   | 4.49 ± 0.31  | 3.96 ± 1.09  | 5.10 ± 1.56   |
| <i>Females</i>                   |                           |              |               |              |              |               |
| <i>Dipodillus dasyurus</i>       | 8.61 ± 0.60               | 9.02 ± 0.31  | 3.27 ± 0.21   | 3.57 ± 0.14  | 2.15 ± 0.89  | 2.22 ± 0.52   |
| <i>Acomys caharinus</i>          | 14.56 ± 0.73              | 15.22 ± 0.79 | 5.88 ± 0.30   | 5.98 ± 0.15  | 3.39 ± 0.61  | 4.1 ± 0.39    |
| <i>Gerbillus pyramidum</i> ; WR  | 25.09 ± 2.78              | 27.08 ± 2.52 | 9.74 ± 1.08   | 10.49 ± 0.98 | 4.59 ± 0.48  | 4.77 ± 0.63   |
| <i>Gerbillus pyramidum</i> ; Sed | 26.28 ± 0.55              | 27.08 ± 2.62 | 10.46 ± 0.47  | 8.68 ± 0.88  | 3.72 ± 0.97  | 5.82 ± 1.43   |
| <i>Gerbillus andersoni</i> ; WR  | 10.56 ± 0.47              | 11.41 ± 0.54 | 4.31 ± 0.14   | 4.51 ± 0.25  | 1.68 ± 0.68  | 1.60 ± 0.23   |
| <i>Gerbillus andersoni</i> ; Sed | 10.26 ± 0.38              | 10.56 ± 0.93 | 4.12 ± 0.14   | 4.19 ± 0.35  | 2.41 ± 0.67  | 3.07 ± 0.74   |
| Group                            | Percent Carcass Component |              |               |              |              |               |
|                                  | %CW                       |              | %FFDM         |              | %BF          |               |
|                                  | LD                        | SD           | LD            | SD           | LD           | SD            |
| <i>Males</i>                     |                           |              |               |              |              |               |
| <i>Dipodillus dasyurus</i>       | 62.11 ± 2.89              | 56.29 ± 3.21 | 24.14 ± 0.29  | 22.92 ± 1.19 | 14.28 ± 3.19 | 20.79 ± 4.32  |
| <i>Acomys caharinus</i>          | 66.32 ± 1.27              | 60.19 ± 0.5* | 24.89 ± 0.34  | 24.71 ± 0.14 | 8.79 ± 1.40  | 15.11 ± 0.47* |
| <i>Gerbillus pyramidum</i> ; WR  | 63.27 ± 2.16              | 64.53 ± 1.31 | 24.98 ± 0.89  | 24.93 ± 0.11 | 11.76 ± 2.93 | 15.18 ± 1.24  |
| <i>Gerbillus pyramidum</i> ; Sed | 61.85 ± 2.45              | 60.91 ± 2.52 | 25.91 ± 3.21  | 24.24 ± 1.83 | 12.23 ± 1.09 | 14.85 ± 4.15  |
| <i>Gerbillus andersoni</i> ; WR  | 60.44 ± 3.34              | 63.65 ± 3.09 | 24.31 ± 0.49  | 26.37 ± 1.27 | 15.25 ± 3.57 | 10.31 ± 4.08  |
| <i>Gerbillus andersoni</i> ; Sed | 58.51 ± 2.76              | 55.41 ± 4.54 | 23.09 ± 1.30  | 22.28 ± 1.5† | 18.40 ± 4.04 | 20.72 ± 5.25† |
| <i>Females</i>                   |                           |              |               |              |              |               |
| <i>Dipodillus dasyurus</i>       | 62.34 ± 2.81              | 61.18 ± 1.70 | 23.74 ± 1.18  | 24.22 ± 1.06 | 13.91 ± 3.98 | 14.59 ± 2.65  |
| <i>Acomys caharinus</i>          | 61.28 ± 1.32              | 60.05 ± 1.45 | 24.76 ± 0.51  | 23.70 ± 0.52 | 13.96 ± 1.71 | 16.25 ± 1.62  |
| <i>Gerbillus pyramidum</i> ; WR  | 63.46 ± 0.55              | 63.86 ± 0.75 | 24.64 ± 0.28  | 24.84 ± 0.63 | 11.89 ± 0.79 | 11.30 ± 1.13  |
| <i>Gerbillus pyramidum</i> ; Sed | 65.29 ± 2.18              | 62.43 ± 0.91 | 25.93 ± 0.70  | 24.73 ± 0.72 | 8.78 ± 2.83  | 12.85 ± 1.52  |
| <i>Gerbillus andersoni</i> ; WR  | 63.83 ± 2.96              | 65.10 ± 0.81 | 26.05 ± 1.09  | 25.76 ± 0.58 | 10.13 ± 4.00 | 8.31 ± 1.25   |
| <i>Gerbillus andersoni</i> ; Sed | 61.34 ± 2.30              | 59.40 ± 2.64 | 24.65 ± 0.99  | 23.43 ± 1.23 | 14.01 ± 3.27 | 15.63 ± 3.46† |

Data represent means ± SE. CW, Carcass water; FFDM, fat-free dry mass; BF, body fat; Percent carcass components calculated as [(carcass component/carcass wet wt) × 100]. \**P* < 0.05 vs. LD-housed animals; †*P* < 0.05 vs. WR animals.

stimulated spermatogenesis, but not testes masses or indexes in both gerbil species compared with their sedentary counterparts and did not affect female reproductive status. Finally, the photoperiod minimally af-

fected body and fat pad masses and carcass composition, except for male *Acomys caharinus*, where IWAT mass and carcass lipid content were increased after SD exposure.

The results of the present study are consistent with studies on the effects of the photoperiod on desert pocket mice (*Perognathus formosus*). In this species, LDs stimulate or maintain reproductive functions. The effect of SDs, however, depends on the environmental conditions (e.g., temperature, humidity) that precede them (24). SDs also do not affect body and epididymal fat pad masses in this species (24).

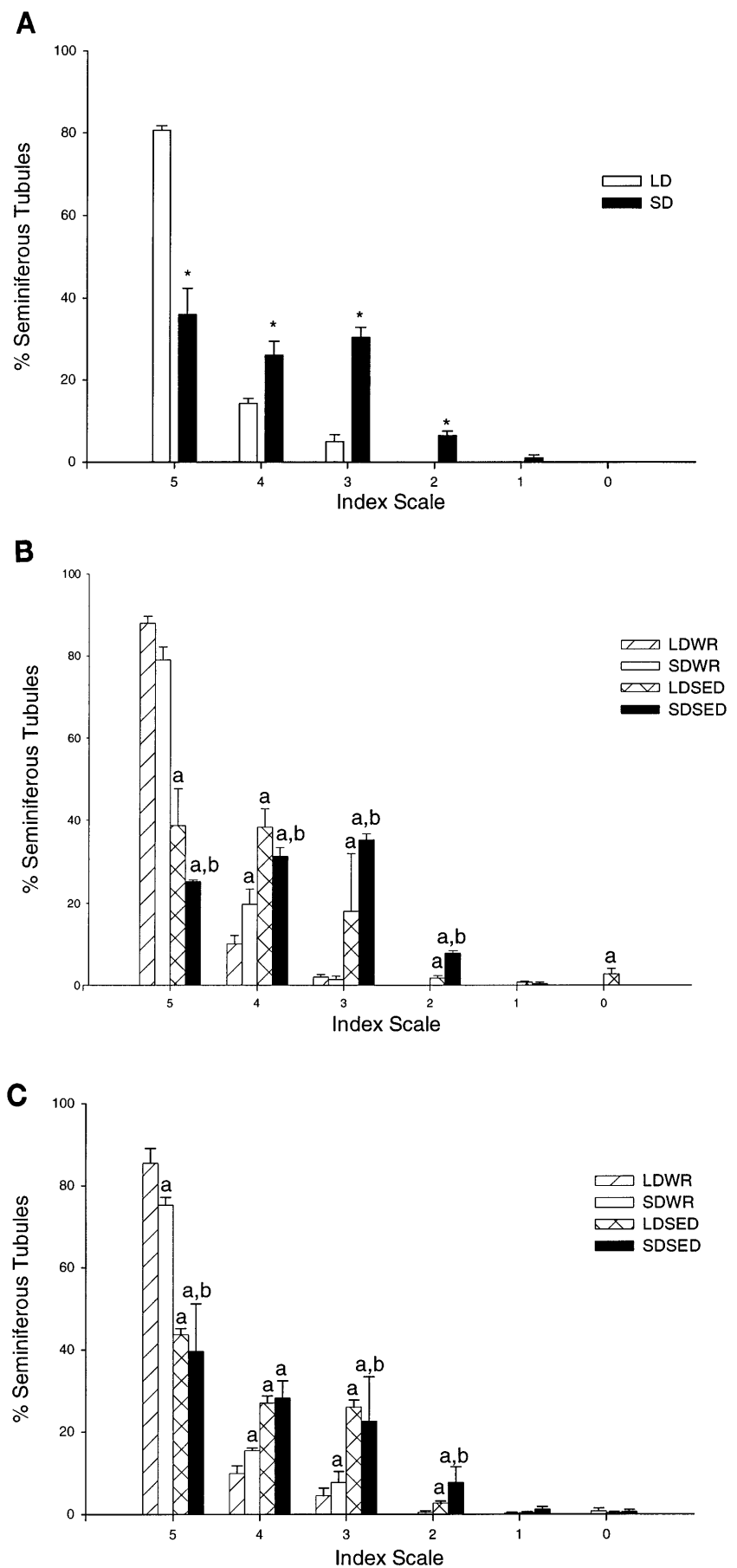
Reproduction in the field is seasonal in the desert rodent species studied in the present experiments, with winter breeding and summer reproductive quiescence (23, 35). Because the animals used in the present experiments were trapped during lengthening days and held in LDs in the laboratory, it is possible that they became photorefractory to the presumed inhibitory effects of LD exposure. Reproductive status was, however, decreased in males of all species after SD exposure; thus it seems unlikely that these animals

Table 3. Effect of photoperiod on testes and uterine masses

| Group                            | Males                 |             | Females         |             |
|----------------------------------|-----------------------|-------------|-----------------|-------------|
|                                  | Paired Testes Mass, g |             | Uterine Mass, g |             |
|                                  | LD                    | SD          | LD              | SD          |
| <i>Dipodillus dasyurus</i>       | 0.07 ± 0.01           | 0.07 ± 0.01 | 0.05 ± 0.02     | 0.08 ± 0.01 |
| <i>Acomys caharinus</i>          | 0.30 ± 0.04           | 0.27 ± 0.01 | 0.07 ± 0.002    | 0.06 ± 0.02 |
| <i>Gerbillus pyramidum</i> ; WR  | 0.77 ± 0.04           | 0.82 ± 0.03 | 0.08 ± 0.02     | 0.13 ± 0.03 |
| <i>Gerbillus pyramidum</i> ; Sed | 0.90 ± 0.09           | 0.78 ± 0.04 | 0.06 ± 0.01     | 0.09 ± 0.04 |
| <i>Gerbillus andersoni</i> ; WR  | 0.61 ± 0.01           | 0.64 ± 0.07 | 0.04 ± 0.01     | 0.10 ± 0.02 |
| <i>Gerbillus andersoni</i> ; Sed | 0.59 ± 0.07           | 0.58 ± 0.06 | 0.04 ± 0.02     | 0.09 ± 0.02 |

Data represent means ± SE.

Fig. 3. Spermatogenic index of male *Acomys cahirinus* (A), *Gerbillus pyramidum* (B), and *Gerbillus andersoni* (C). LD, long photoperiod; LDWR, LD-housed wheel running animals; SDWR, SD-housed wheel running animals; LDSED, LD-housed sedentary animals; SDSED, SD-housed sedentary animals. \* $P < 0.05$  vs. LD-housed animals; <sup>a</sup> $P < 0.05$  vs. LD-housed wheel running animals; <sup>b</sup> $P < 0.05$  vs. LD-housed sedentary animals. See MATERIALS AND METHODS for details.



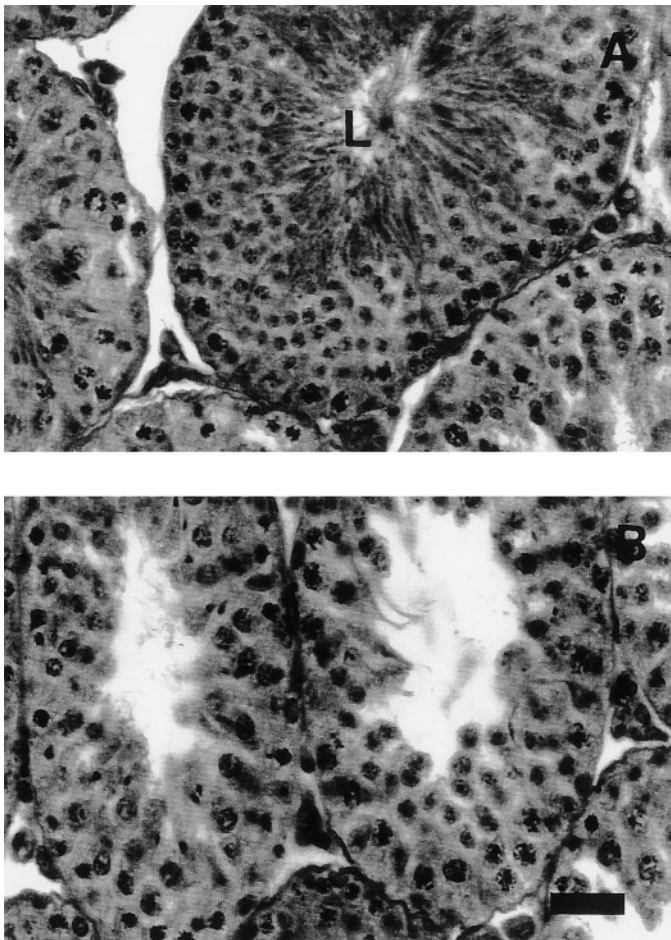


Fig. 4. Testis sections of *Dipodillus daysurus*. Spermatogenically active testes from LD-housed animal (A) and SD-housed animal (B); all stages of spermatogenesis are present with the tails of mature spermatozoa extending into the lumen of the seminiferous tubule (L). Note reduced spermatogenesis in SD-housed animal. Bar, 50  $\mu$ m.

were photorefractory. The absence of an inhibitory effect of LDs on the reproductive status of males and females of all species suggests that the seasonal reproductive patterns of these rodents may not be photoperiodically regulated in the wild. Nevertheless, the inhibitory effect of SDs on male reproductive status indicates that these animals can discriminate SDs from LDs. When the possible relations between tropical mammals and photoperiodism (20) are extended to these and perhaps other desert rodents, it would appear that they are moderately photoresponsive but that they may not use day length as the primary proximal cue to trigger reproductive and other seasonal responses. Although the annual variation in day lengths in Egypt is sufficient to permit dependence on the photoperiod as a proximate factor to regulate reproduction and other seasonal changes (5), it might not be expected that desert rodents depend on the photoperiod to cue their reproduction given the harsh, unpredictable climate of the desert (7). Therefore, a possible explanation for the present results is that photoperiodic inhibition of reproduction may be overridden by other factors in their natural habitat that allow breeding to coincide with the

rainy season when food is abundant. This suggestion parallels findings on California voles (*Microtus californicus*), another winter-breeding species (33). In the laboratory, male California voles undergo gonadal regression in SDs, but the ingestion of green vegetation overrides the inhibitory influence of SD exposure. Furthermore, it is possible that the inhibitory effect of SDs reported in the present investigation may be overridden in the natural habitat by social cues. The role of social cues in regulating reproduction has been observed in many rodent species (for review, see Ref. 7). For example, the presence of females enhances male reproductive functions under LD conditions in some species (12) and overrides the inhibitory reproductive effects of SD exposure in other species (45, 46).

In the present study there was individual variation in the rates of testicular regression within the same species and between different species. Such variation in reproductive responsiveness to SDs has been seen in many other species and is thought to be genetically based (13, 21, 32) but may be environmentally triggered (17).

Testes of animals with access to running wheels had significantly increased spermatogenic activity compared with their sedentary counterparts. These results are consistent with stimulation of reproductive functions of other photoperiodic and nonphotoperiodic rodent species by exercise (8, 26, 27, 37, 46). The underlying mechanism for this effect is not precisely known, but the hypothalamic-pituitary-gonadal axis may be involved (26, 27). Although the changes in spermatogenic activity reported in the present study between wheel running and sedentary animals were not robust, it may be significant to the animals in the wild and could indicate the possibility that locomotor activity antagonizes the inhibitory effect of SDs on reproductive functions.

In many rodent species, the photoperiod affects female and male reproductive status similarly. For example, exposure of male Siberian hamsters to SDs decreases testes and uterine masses as well as eliminates ovarian corpora lutea and antral follicles (43, 44). This was not found in the present study, where the ovaries of LD- and SD-housed animals of all species exhibited well-developed antral follicles and corpora lutea, although their male counterparts underwent partial gonadal regression. It is not known why females

Table 4. Effect of photoperiod on testicular indexes

| Group                            | Testicular Index |                 |                  |
|----------------------------------|------------------|-----------------|------------------|
|                                  | LD               |                 | SD               |
|                                  | T (same)         | T (diff)        |                  |
| <i>Gerbillus pyramidum</i> ; WR  | 1.33 $\pm$ 0.06  | 1.02 $\pm$ 0.03 | 0.82 $\pm$ 0.03* |
| <i>Gerbillus pyramidum</i> ; Sed | 1.20 $\pm$ 0.04  | 0.90 $\pm$ 0.05 | 0.77 $\pm$ 0.04* |
| <i>Gerbillus andersoni</i> ; WR  | 0.93 $\pm$ 0.05  | 0.8 $\pm$ 0.06  | 0.67 $\pm$ 0.05* |
| <i>Gerbillus andersoni</i> ; Sed | 1.1 $\pm$ 0.04   | 0.89 $\pm$ 0.07 | 0.67 $\pm$ 0.05* |

Data represent means  $\pm$  SE. T (same), testicular indexes measured in the same animals before transferring them to SD; T (diff), testicular indexes measured in different animals. \* $P$  < 0.05 vs. LD-housed animals.



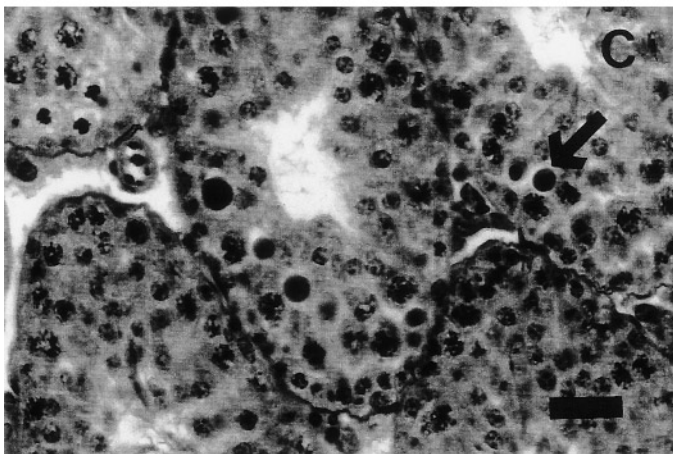
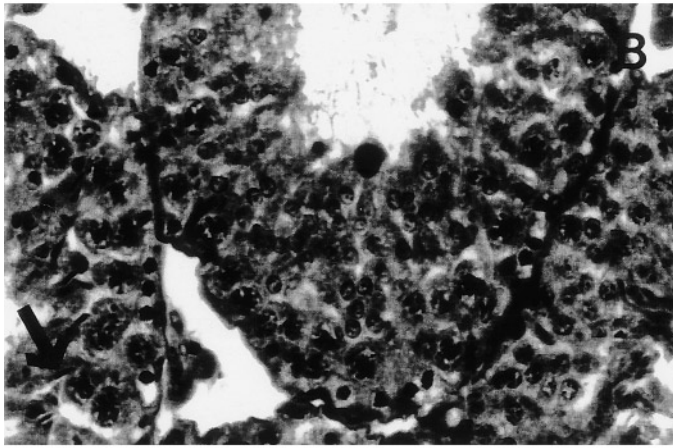
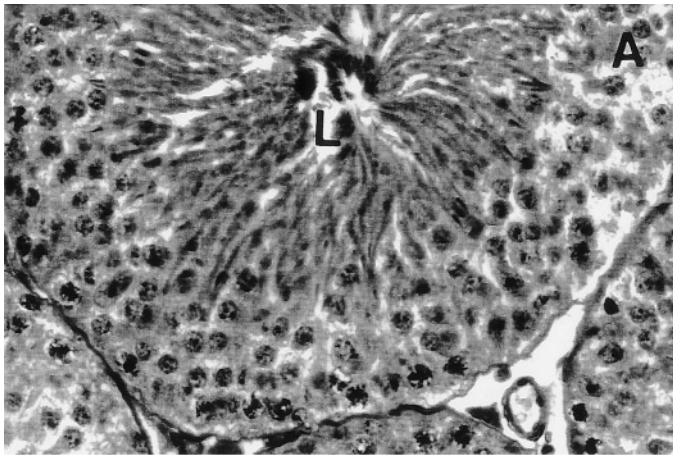


Fig. 5. Testis sections of *Gerbillus andersoni* from LD-housed animal (A), illustrating full spermatogenic activity with the tails of mature spermatozoa extending into the lumen of the seminiferous tubule and from SD-housed wheel running animal (B), showing less spermatogenic activity. Note that some seminiferous tubules still have maturing spermatozoa (arrow). C is from SD-housed sedentary animal, showing severe testicular regression. Note absence of late stages of spermatogenesis (long spermatids and spermatozoa), presence of many necrotic cells (arrow), and the reduced diameter of the seminiferous tubules. All sections are at the same magnification. Bar, 50  $\mu$ m.

in these species differ from males in lacking even mild reproductive photore sponsiveness.

In summary, the present study demonstrates that some desert rodent species are responsive to the photoperiod but that changes in the photoperiod alone are not enough to induce robust changes in reproductive responses. Therefore, it is suggested that alterations in the day length cannot, in and of themselves, account for the seasonal reproductive cycles seen by these species in the wild.

#### Perspectives

The environmental factors that regulate the reproductive patterns of desert rodents in the wild remain to be identified. There is some evidence that desert rodents generally curtail breeding during the dry season and breed only during the rainy season in the following months when desert grasses are abundant (19). One possible outcome of this scenario is that desert rodents use the ingestion of green food for timing their reproduction (for review, see Ref. 7). Reliance on a secondary plant compound (e.g., 6-methoxy-2-benzoxazolinone) found in young grasses and sedges to predict oncoming periods of maximum food availability, and thus conditions favorable for reproduction, has been documented in other species of small mammals (1, 42). Desert rodents could use such plant predictors in timing their reproduction with or without photoperiodic regulation (7). Strict reliance on the photoperiod as a predictor for timing the reproductive season, however, could be

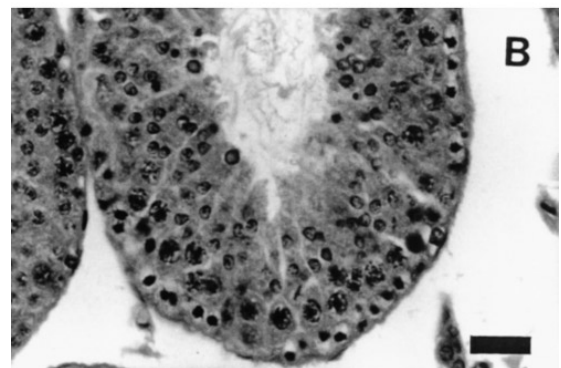
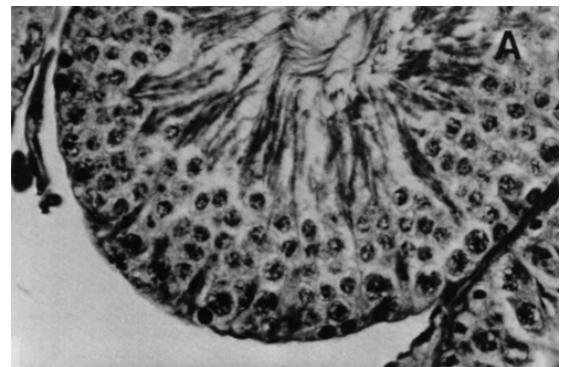


Fig. 6. Testis sections of *Gerbillus pyramidum*. Spermatogenically active testis are from sedentary LD-housed animal (A), illustrating full spermatogenic activity and from SD-housed sedentary animal (B), showing reduced spermatogenesis. Bar, 50  $\mu$ m.

disadvantageous in desert environments in which climatic and nutritional conditions change in an unpredictable manner (10). Because of the harsh desert environment, reproduction in desert species may not be successful in any given year (25), suggesting that desert rodents breed opportunistically depending on relatively short-term climatic and nutritional conditions without using any predictors. In addition, they also may have overcome some of their reproductive difficulties by being long-lived. Although there are no available field data, this may be the case for our animals. For example, longevity of a captive specimen of *Gerbillus pyramidum* was 8.17 yr (reviewed in Ref. 34). Taken together, the results of the present study suggest that photoperiod may play a role in seasonal breeding in some desert rodents, although other environmental factors also may contribute to the seasonal breeding seen in the wild.

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